MnPASS System
Today and the Future

April 2010

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Minnesota Department of Transportation

Your Destination...Our Priority

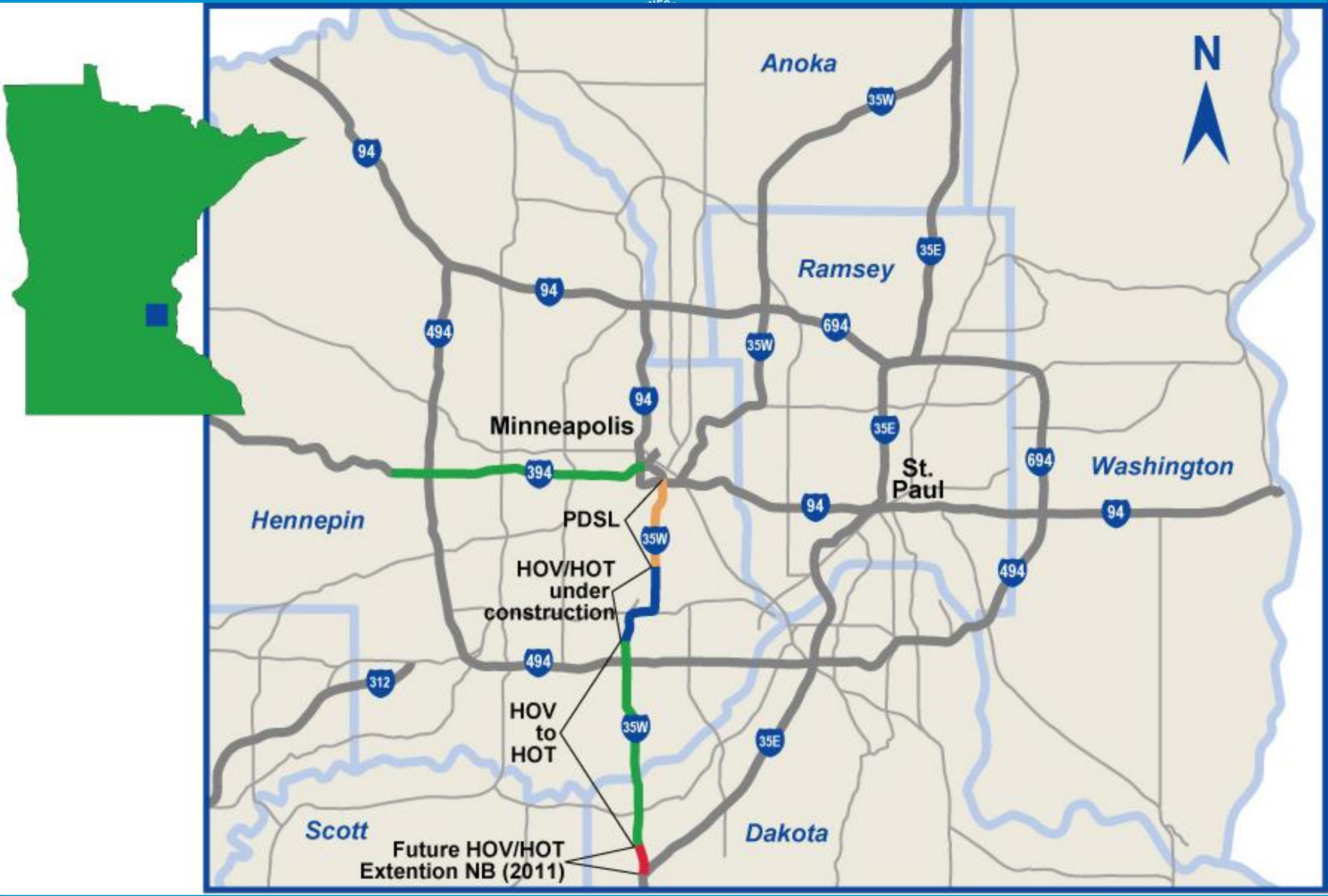
Topics

- **Minnesota's Current and Future MnPASS High Occupancy Toll Systems**
  - **Overview of the MnPASS System**
  - **Update on past and present studies**

Congestion Pricing In Minnesota

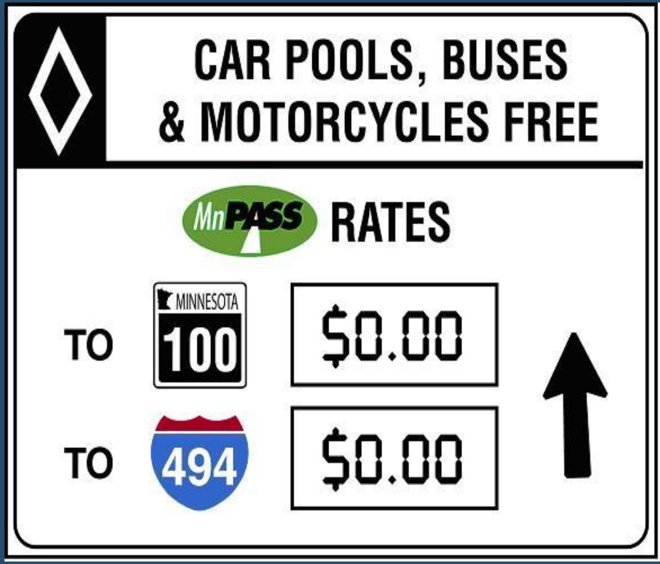
- **Opened 11 mile High Occupancy Toll (HOT) lane on I-394 in 2005**
- **Opened new 16 mile HOT lane in phases on I-35W**
  - **8 miles of HOT lanes opened in 2009**
  - **3 miles of Priced Dynamic Shoulder Lanes**
  - **4 miles of HOT to open in 2010**
- **Congestion pricing brand name:**

MnPASS


Anoka
Ramsey
Washington
Hennepin
Scott
Dakota
Minneapolis
St. Paul
N
94
494
35W
694
35E
394
312
PDSL
HOV/HOT under construction
HOV to HOT
Future HOV/HOT Extention NB (2011)

Electronic Toll Collection

- **Min: $0.25 Max $8.00**
  - **I-394: 90% pay $3 or less per trip**
  - **I-35W: 90% pay $1.75 or less per trip**
- **Price based on MnPASS lane traffic conditions**
- **Data from traffic sensors in lane**
- **Price updated every 3 minutes**
- **Pricing during set hours only (i.e. not 24x7)**



Technology Provided to Enforce MnPASS

MnPASS Users Satisfied

- 91% overall satisfaction
- 95% satisfaction with all electronic tolling
- 85% satisfaction with traffic speed in lane
- 76% satisfaction with dynamic pricing
- 66% satisfaction with safety of merging

Corridor Operations Improve with MnPASS

- I-394 has 5,000 MnPASS customers per day; avg. toll $1.15
- I-394 MnPASS lanes peak hour volumes increased 9 to 33%
- Total I-394 peak hour roadway volumes increased by up to 5%
- 98% of time speeds above 50 mph
- Travel speeds in the regular lanes increased by 2 to 15%
- Transit ridership and carpools levels increase
- 45% reduction in crashes

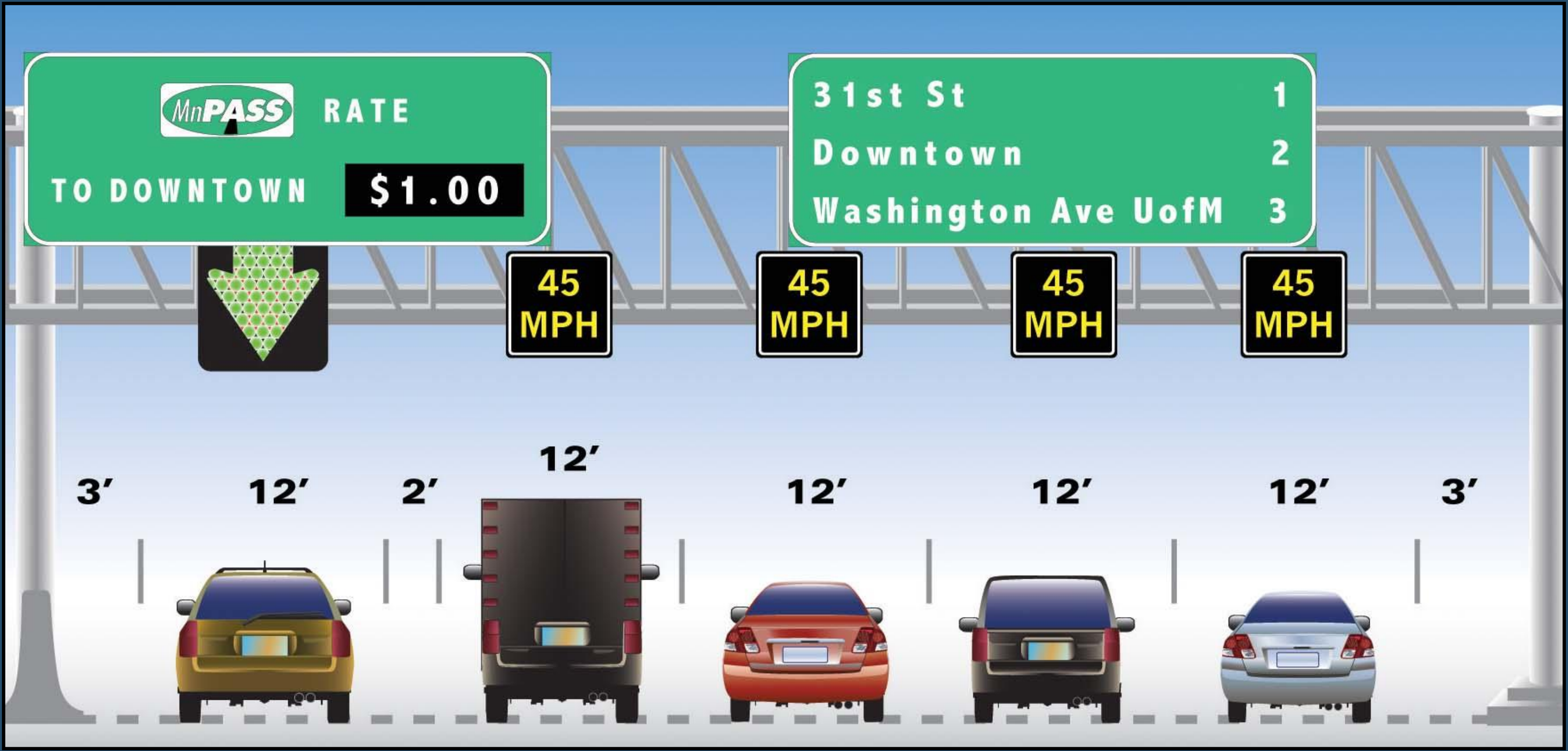
I-35W MnPASS Early Results

- 1200-1500 MnPASS users per day
- 51% stay on I-35W north of I-494 (2 pricing zones)
- 49 % exit at or before I-494
- 4800 transponders holders in I-35W corridor
- 1400 new transponder holders in I-394 corridor
  – Marketing created spill over demand
- Substantially more use expected on I-35W once Crosstown Commons area is complete in 2010

I-35W MnPASS: Active Traffic Management

I-35W MnPASS: Compact Design

35W MnPASS on Shoulder: A New Design Option



I-35W MnPASS on Shoulder

I-35W MnPASS on Shoulder: Technology is Key

What's Next for MnPASS?

- By Fall 2010 all existing HOV lanes converted to HOT
- Seeking MnPASS expansion opportunities under several design concepts:
  - 2 miles extension of I-35W HOT lane in 2011
  - Opening dynamic shoulder lanes and ATM on 10 mile I-94 Corridor in 2011 with future MnPASS
- MnPASS Next: Planning Phase
  - TH 77 MnPASS+ Study complete in 2010
  - MnPASS Network Study to identify candidate corridors for expansion of MnPASS into a regional network

MnPASS System Planning Studies

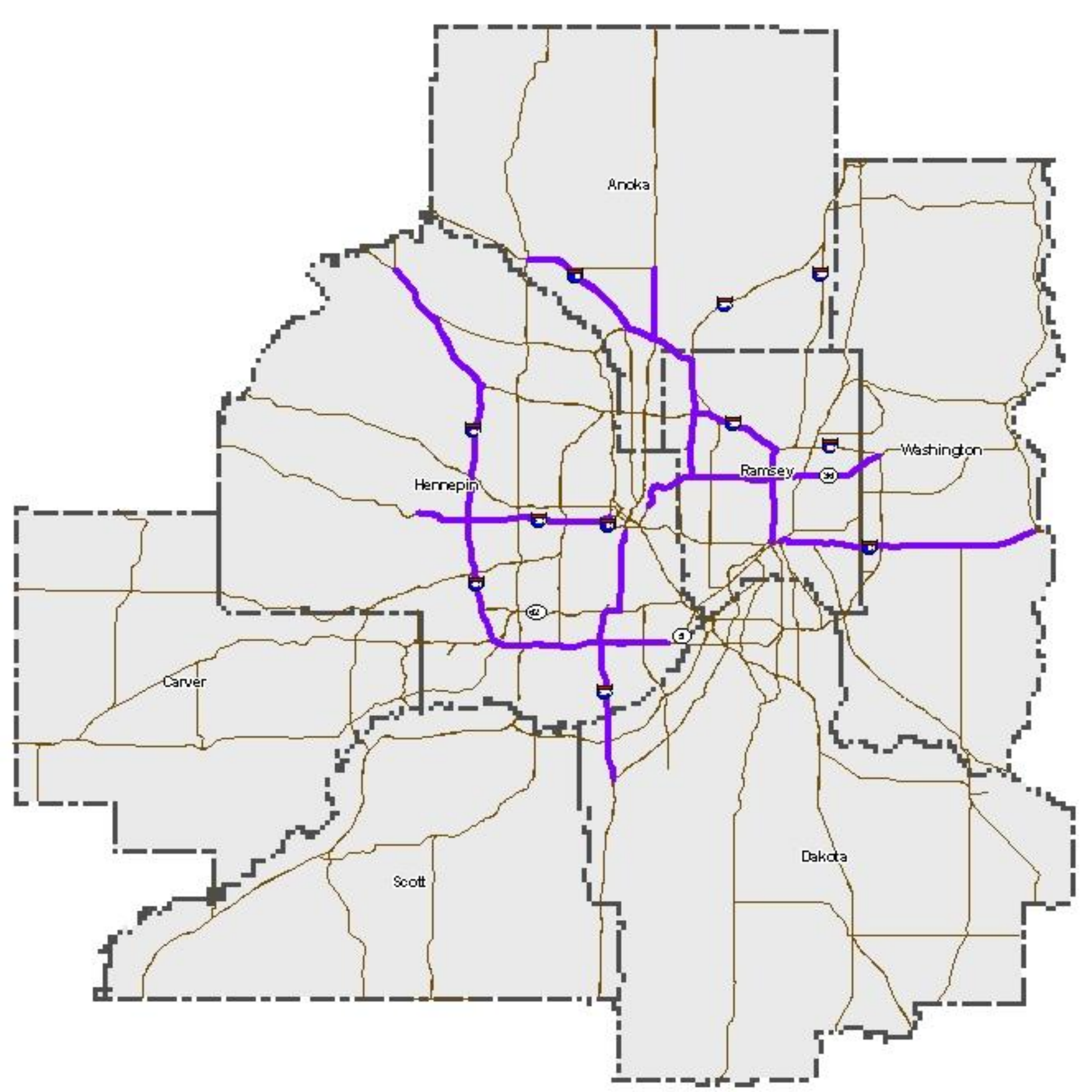
MnPASS System Study #1 (2005)

- A technical analysis to identify a potentially viable system of MnPASS lanes and associated benefits
  - Included one round of initial screening and two rounds of increasingly detailed analysis
- A road map to implementation and prioritization
  - Identify short and long-term opportunities
- An identification of implementation issues
  - Capital and operating costs, revenue potential
  - Travel benefits
  - Operational considerations
  - Impacts on existing transportation system and policy plans

System Study 1: Emerging Vision

- **An interconnected *system* of uncongested, variably priced toll lanes**
  - **Electronic toll collection, no toll booths**
    - **Variable message signs advise drivers of toll rates**
- **Priced to help manage congestion**
  - **Speeds at or near posted limits maintained by pricing that varies with demand and use**
- **Transit friendly** (provide opportunities for transit enhancements)
- **Another source of transportation funding** (but do not have to pay their own way)

MnPASS System Vision in 2005



Acknowledged capacity needs not otherwise addressed in the TPP

Includes all of Concept A minus the recently completed stretch of I-694 and a piece of TH36 (from the beltway east)

Includes a fourth lane on I-494 between I-35W and I-394 (extended to TH 5)

Mapped against and found to be consistent with the regional models population and economic growth assumptions

2005 Summary of Findings

- MnPASS lanes are a new transportation “product” that can provide desired impacts
  - Significant time savings for MnPASS users over non-users (a congestion-free choice/alternative)
    - Time savings alone leads to 1.26 benefit/cost ratio (for Concept A)
  - Non-users also expected to benefit
- Transit systems can benefit
  - Modeling MnPASS express bus service on TH 36 predicted a 6.2% increase in transit ridership
  - BRT has design implications on access to MnPASS lanes, placement of transit stations, and operating speeds

2005 Summary of Findings

- Public investment is required
  - Only 15% to 55% cost recovery
  - Need source of public investment to close funding gap
- Findings conflict with existing plans
  - “Best” potential projects already underway or committed, and are not eligible under current policy
    - e.g. I-494 design build project
  - Other MnPASS segments projected to experience the highest congestion and yield the highest revenues are not in the 30-year Transportation Policy Plan

2010 Studies- Parallel Studies with a new vision

- **Metropolitan Highway System Investment Study (MSHIS)**

- **MnPass System Study II**

MSHIS

- Develop a 50-year Vision for the metropolitan highway system that allows the region to look beyond the fiscally constrained 2030 TPP, Mn/DOT Statewide and District Plans
- Identify investment alternatives to improve metropolitan highway system performance and preserve mobility

MnPASS System Study 2

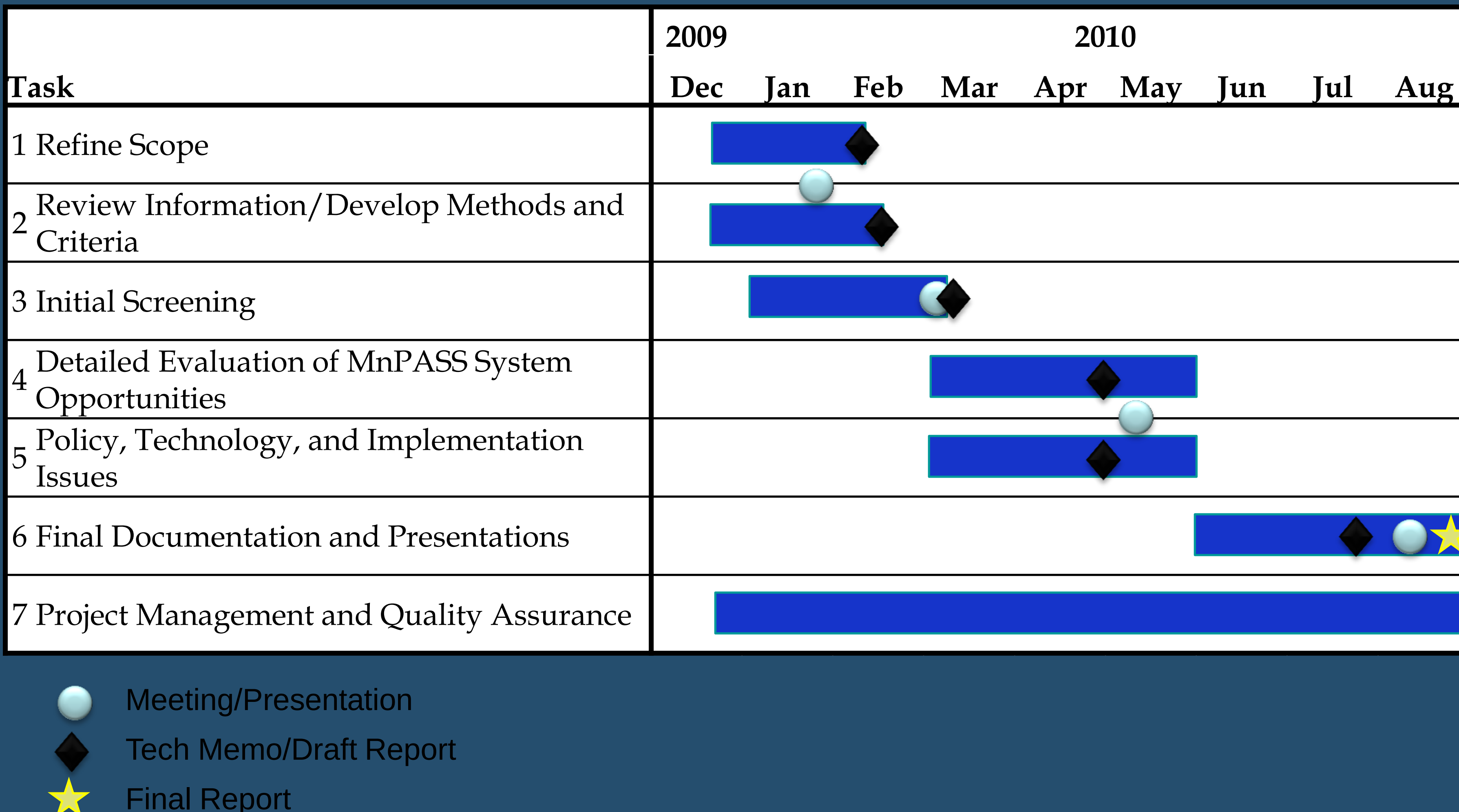
Study Goals

- Identify and prioritize future MnPASS expansion projects from technical standpoint
- Look only at expansion to system capacity
- Utilize new design options to study lower cost options then Study 1 considered
- Coordinate closely with MHSIS Directions

MnPASS System Study 2 Tasks

- Develop criteria to identify viable MnPASS projects
- Develop a prioritized list of potential candidate MnPASS lanes that can be implemented in the near term (2-10 years)
- Conduct traffic and revenue analysis of most promising projects using the Met Council regional model
- Identify the technological, policy, financial and institutional issues and barriers that will need to be addressed to ensure successful implementation of the recommended system.

Study Schedule



Potential MnPASS Corridors

1. TH 36: I-35W to I-694
2. I-94: TH 101 to I-494
3. I-35E: I-94 to CR E
4. I-35W: Minneapolis to Blaine
5. I-494: TH 212 to I-94
6. TH 169: TH 101 to I-94
7. TH 77: 141st Street to TH 62
8. I-94: Downtown Minneapolis to Downtown Saint Paul
9. I-394: TH 100 to I-94
10. I-494: TH 212 to MSP Airport
11. TH 212/TH 62: TH 5 to TH 77
12. I-94: Downtown Saint Paul to I-694
13. TH 280: I-94 to I-35W

MnPASS Lane Configuration- 3 options

- Add full lane on inside

  (4' left shoulder, 14 foot HOT, 12 MUL, 10 Right shoulder)
- Convert shoulder (PDSL) + emergency pullouts
- Use minimum design (2 ft clearances, 11 foot lanes)

Construction Feasibility Assessment Factors

- Bridges/Structures (overpasses, underpasses)
- Mainline (centerline spacing - 38.5ft minimum)
- Interchange modifications
- Major connection issues

Screening Categories

- **Group 1 - Most feasible** (least amount of construction in scope and magnitude; likely to meet standard geometry)
- **Group 2 – Challenging** (Could be done with reconstruction; may require alternative design to minimize cost and impacts)
- **Group 3 - Most difficult** (significant bridge, mainline, interchange and connection issues; may need to look at design exceptions other innovations to make work)

6

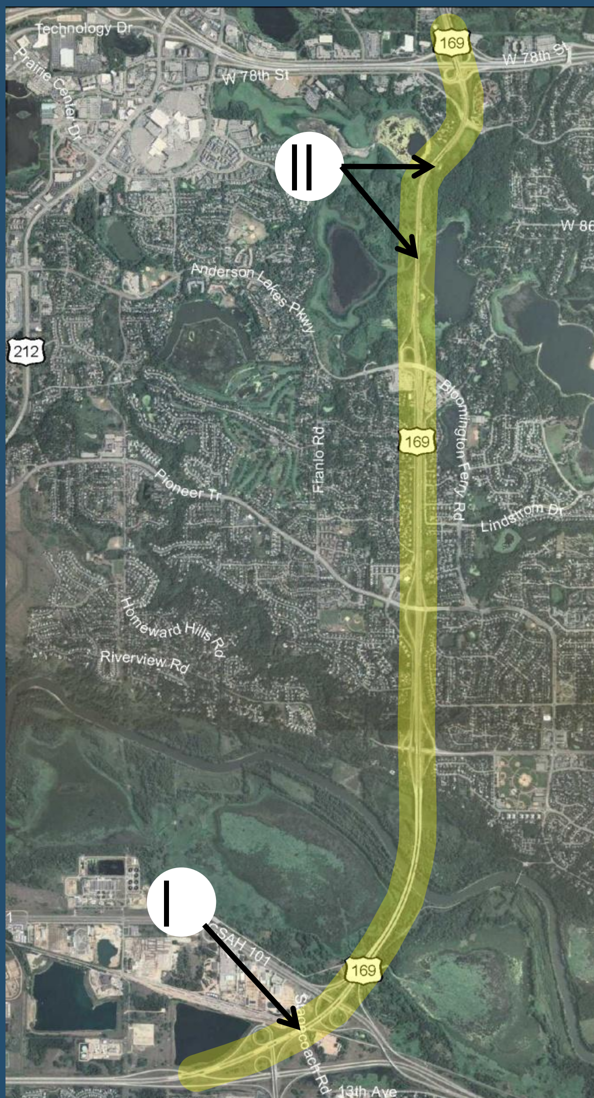
TH 169: TH 101 to I-94
3 Segments Considered

- **4 to 6 General Purpose Lanes, Auxiliary Lanes**
- **Traffic Volumes**
  - **Segment A: 65,000 to 80,000 vpd**
  - **Segment B: 55,000 to 100,000 vpd**
  - **Segment C: 80,000 to 100,000 vpd**
- **Transit Facilities - Bus Shoulders**
- **Congestion Issues**
  - **AM Peak: 2-3 hours**
  - **PM Peak: 2-3 hours**

6a

TH 169: TH 101 to I-94

TH 101 to I-494 (South boundary changed to CR 17)



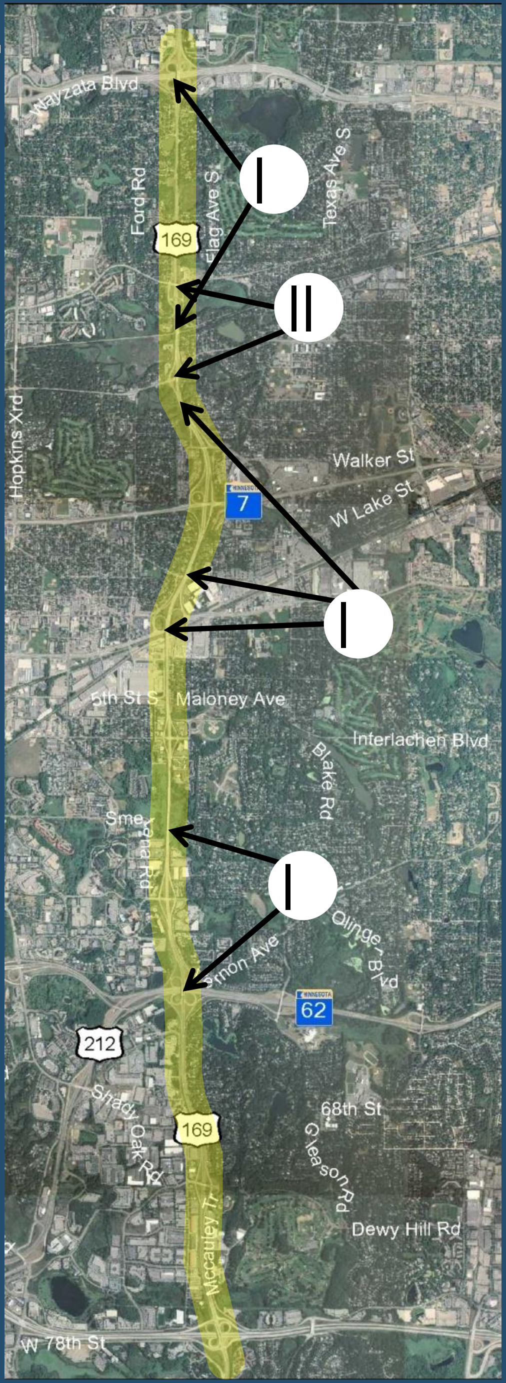
I. **Start/end MnPass lanes between TH 13 connections and CSAH 18; end 3rd westbound lane at CSAH 83**
II. **Widen two bridges over Anderson Lakes**

6b TH 169: TH 101 to I-94

I-494 to I-394

I. **Widen bridges over TH 62/TH 212, Nine-Mile Creek, Excelsior Blvd, Minnetonka Mills, Minnehaha Creek, railroad, I-394 Frontage Rd**

II. **Replace bridges at Minnetonka Blvd and Cedar Lake Rd**

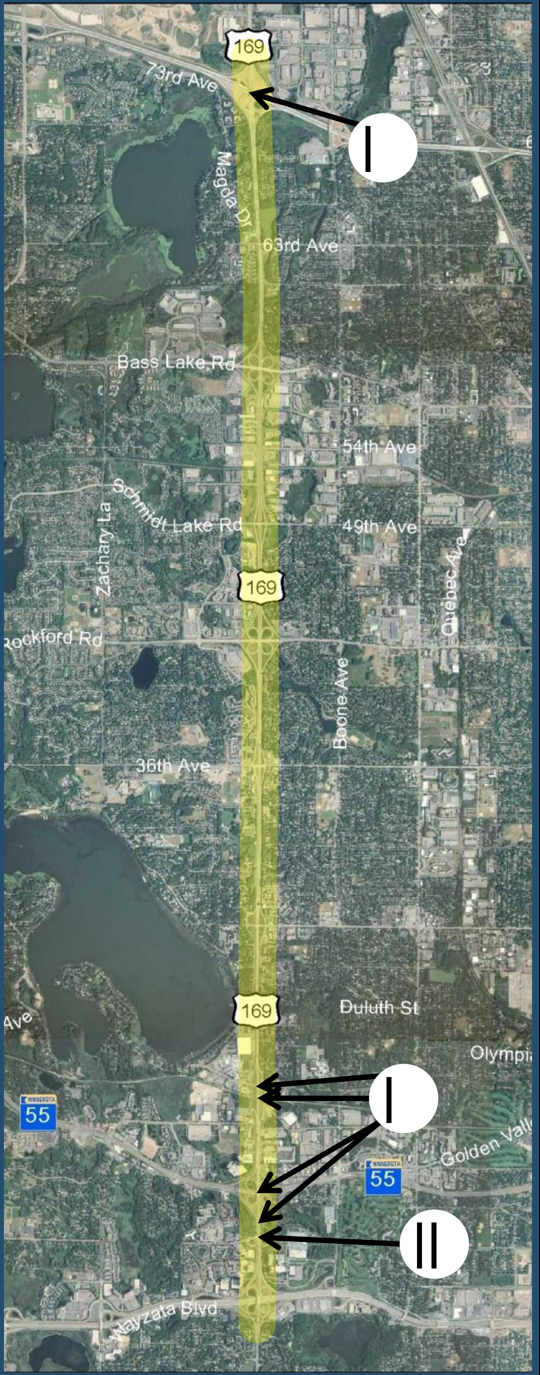


6c TH 169: TH 101 to I-94

I-394 to I-94

I. **Widen bridges over Willow Creek, TH 55, railroad, Plymouth Ave, I-94**

II. **Replace Betty Crocker Dr bridge**



Status of Hwy169 Segments

- **Segment A: Continues to be studied**
  - Segment boundaries extended to CR 17
  - Issues are at bridge and interchange south of river
- **Segment B: Dropped from further study**
  - Technically complicated (i.e. costly) due to substantial bridge widening needs
- **Segment C: Dropped from further study**
  - Parallel corridor (5B) more favorable option

Remaining Steps

- **Complete technical analysis**
- **Identify policy issues**
- **Incorporate MnPASS designs for high priority corridors into planned projects**

More Information:

Visit

www.mnpass.org

Contact

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